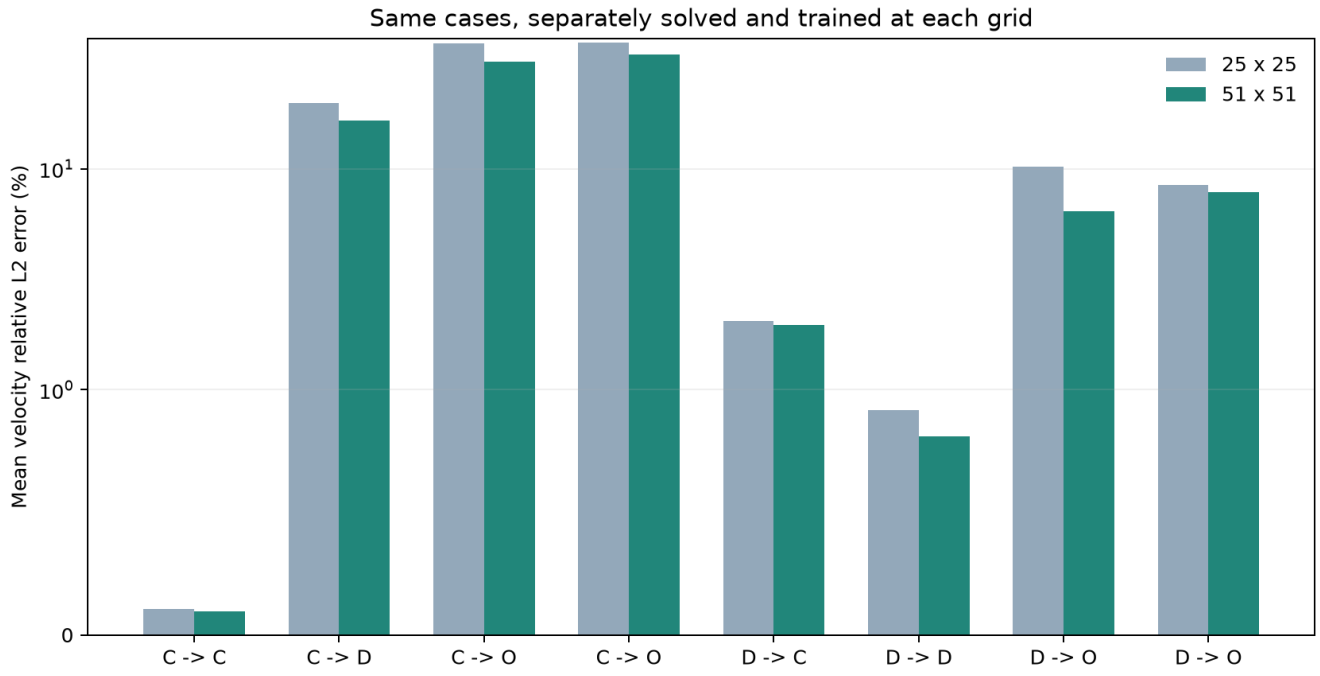
51 x 51 numerical refinement

Matched Stokes fields and Navier-Stokes labels were solved again on 51 x 51 nodes.

The retained case manifest is unchanged; tests are regression cases, not fresh blind data.

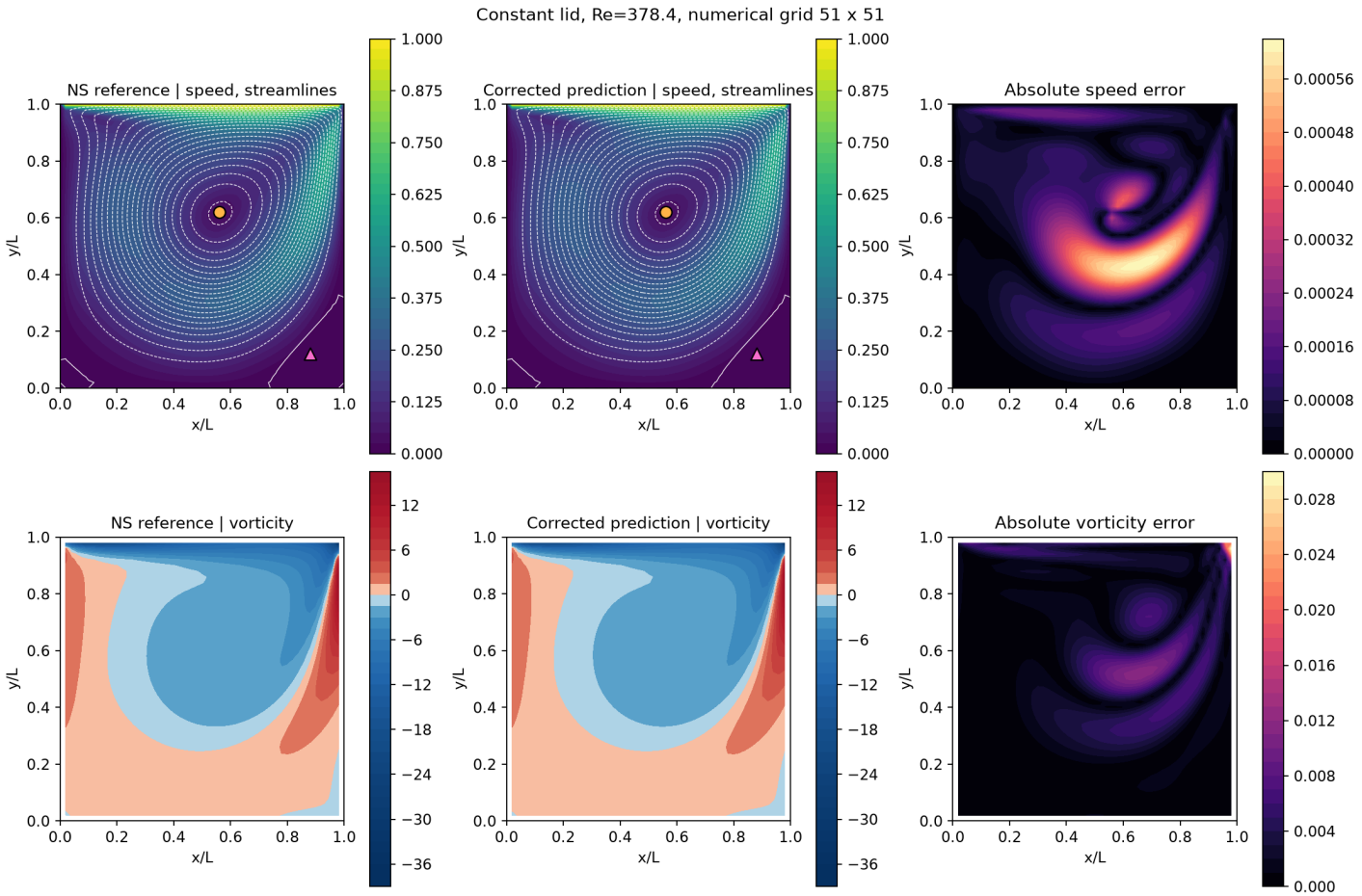
The selected POD-MLP architecture was retrained from scratch at 51 x 51.



Train -> test	**25 x 25**	**51 x 51**
constant -> constant	0.107%	0.097%
constant -> diverse	19.920%	16.555%
constant -> ood_re	36.982%	30.700%
constant -> ood_shape	37.192%	33.068%
diverse -> constant	2.044%	1.960%
diverse -> diverse	0.917%	0.810%
diverse -> ood_re	10.204%	6.425%
diverse -> ood_shape	8.428%	7.878%

Errors compare models with separate numerical labels on each grid.

Constant lid: speed, streamlines, vorticity



Mean CFD 25-to-51 velocity change at coarse nodes: 20.59%

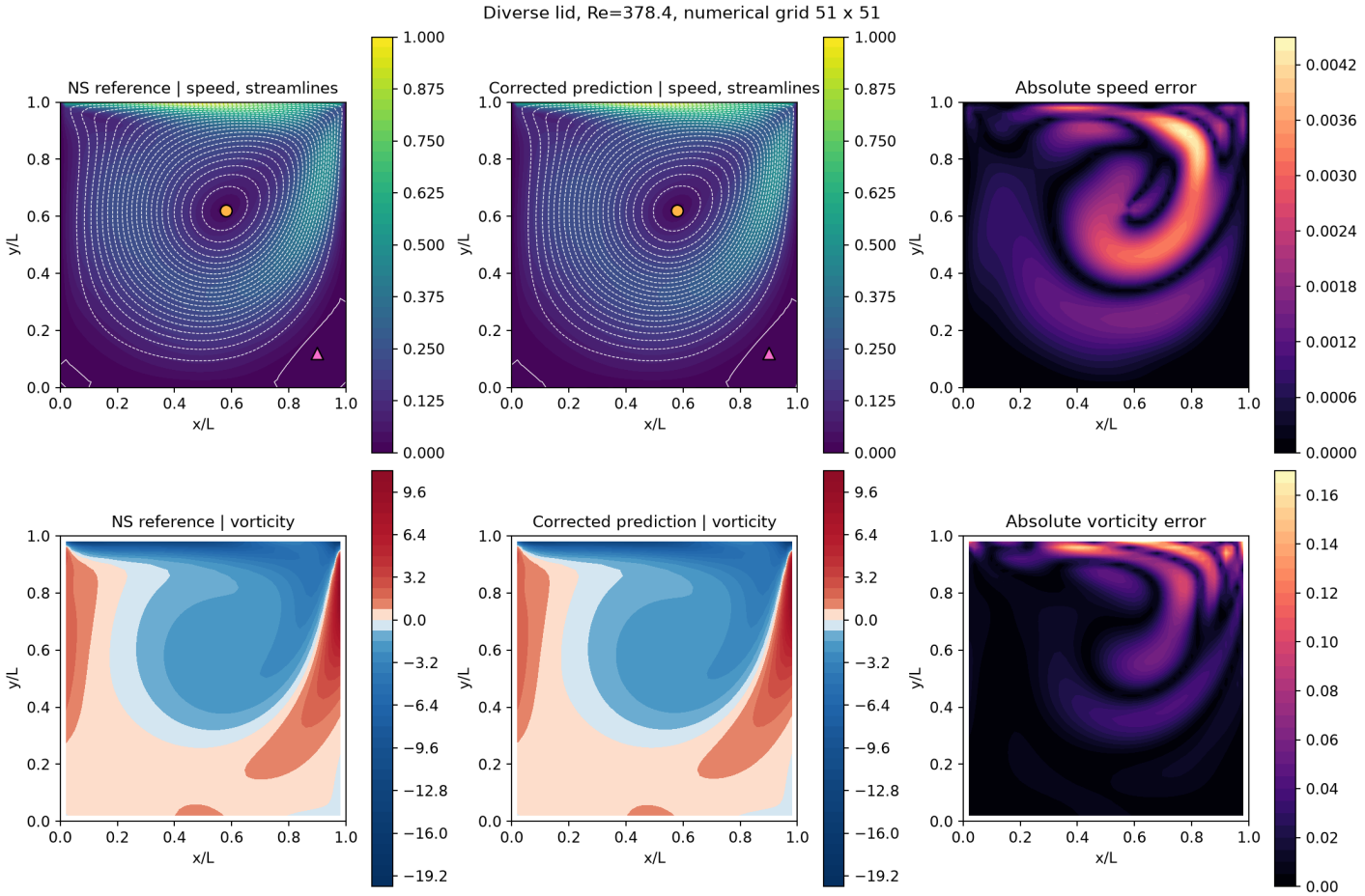
Vortex positions are measured on 51-node arrays; display contours do not add resolution.

One refinement step does not establish mesh-independent vortex capture.

Inter-grid differences are not estimates of either grid's true error without further assumptions.

Small same-grid surrogate error establishes label fidelity, not independent physical accuracy.

Diverse lid: speed, streamlines, vorticity



Mean CFD 25-to-51 velocity change at coarse nodes: 17.81%

Vortex positions are measured on 51-node arrays; display contours do not add resolution.

One refinement step does not establish mesh-independent vortex capture.

Inter-grid differences are not estimates of either grid's true error without further assumptions.

Small same-grid surrogate error establishes label fidelity, not independent physical accuracy.

Week 4.2 | Reading the evidence

Separate surrogate error from changing the CFD labels

The low-fidelity Stokes field omits convection; the learned correction estimates what must be added to approximate Navier–Stokes on the specified grid. The surrogate-to-label error and the coarse-to-fine CFD difference answer different questions. A 0.1% surrogate error against a coarse field can coexist with a 20% change in that field under refinement.

Pressure is reconstructed from momentum gradients with an aligned gauge. The reported gradient residual measures compatibility of that reconstruction with the supplied gradients; it is not a direct pressure-error norm and need not rank models in the same order. Read velocity, pressure and compatibility together. Optional retraining writes a new student directory. The retained figures remain historical evidence; they do not become figures of your new fit.

Check your understanding. Identify the input information, the reference, the metric denominator and the evaluation split in one worked notebook comparison. Predict which conclusion would change if any one of those four changed, then check your prediction.